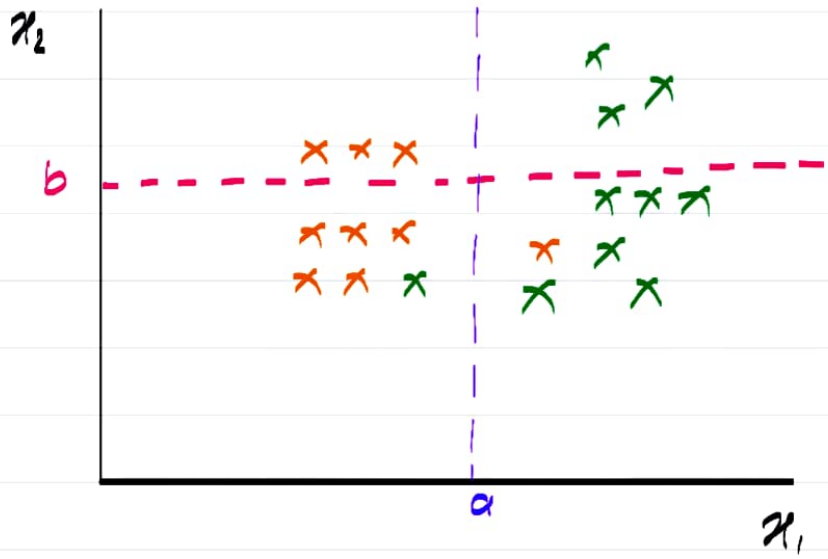
Machine Learning

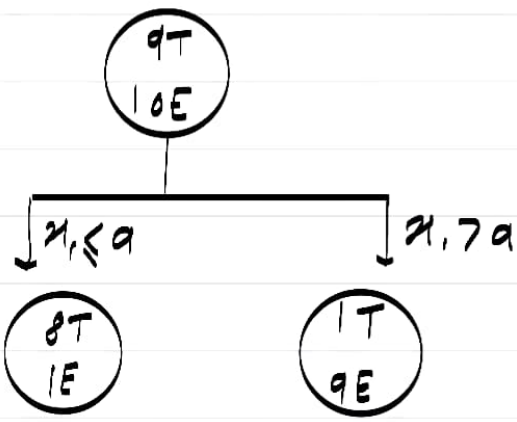
23-04-26

Decision Tree Clf:



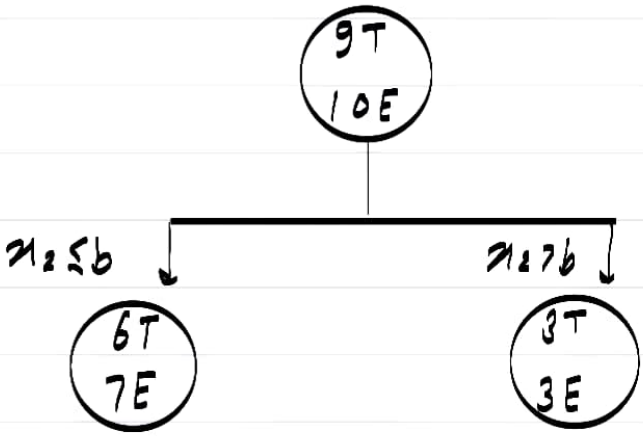
decision rule 1

$x_1 \leq a$



decision rule 2

$x_2 \leq b$



which decision rule is giving purer child nodes?

→ decision rule 1

We find best decision rule using <u>information gain</u>

The rule giving maximum information gain is the best decision rule.

Information gain = entropy of parent node − weighted entropy of child nodes

= entropy before split − entropy after split

$$IG = H(\text{parent}) - \sum H(\text{child nodes})$$

for decision rule 1

$$H(\text{parent}) = -\sum_{i=1}^{2} p_i \log(p_i)$$

$$= -\left[\frac{9}{19} \times \log\left(\frac{9}{19}\right) + \frac{10}{19} \times \log\left(\frac{10}{19}\right)\right]$$

$$= 0.299$$

$$H(\text{left child}) = -\left[\frac{8}{9} \times \log\left(\frac{8}{9}\right) + \frac{1}{9} \times \log\left(\frac{1}{9}\right)\right]$$

$$= 0.151$$

$$H(\text{right child}) = -\left[\frac{1}{10} \times \log\left(\frac{1}{10}\right) + \frac{9}{10} \times \log\left(\frac{9}{10}\right)\right]$$

$$= 0.141$$

$$IG = H(\text{parent}) - \sum_{i=1}^{2} \frac{n(\text{child } i)}{n(\text{parent})} \times H(\text{child } i)$$

$$= 0.299 - \left[\frac{9}{19} \times 0.151 + \frac{10}{19} \times 0.141\right]$$

$$\boxed{IG = 0.1588}$$

for decision rule 2

$$H(\text{left child}) = -\left[\frac{6}{13} \times \log\left(\frac{6}{13}\right) + \frac{7}{13} \times \log\left(\frac{7}{13}\right)\right]$$

$$= 0.299$$

$$H(\text{right child}) = -\left[\frac{3}{6} \times \log\left(\frac{3}{6}\right) + \frac{3}{6} \times \log\left(\frac{3}{6}\right)\right]$$

$$= 0.301$$

$$IG = 0.299 - \left[\frac{13}{19} \times 0.299 + \frac{6}{19} \times 0.301\right]$$

$$\boxed{IG = 0.006}$$

$IG_1 = 0.1588$ $\quad$ $IG_2 = 0.006$

$IG_1 > IG_2$

as $IG_1 > IG_2$, decision rule 1 is better than rule 2

Steps in DT clf training

we have dataset with 'n' features

lets say $n = 3$ features — x_1 x_2 x_3

train set - 80 datapoints

test set - 20 datapoints

target → 3 classes — C_1 C_2 C_3

1) start at the root node, containg all train datapoints

30 C_1
30 C_2
20 C_3

2) set the hyperparameters

criterion : entropy or gini index
max depth : an integer value
min samples split : int
min samples leaf : int
max features : int = 2

3) Start building tree

x_1	x_2	x_3	target
2.5	a	10	C_1
3.7	b	11	C_1
4.5	c	12.2	C_2
⋮	a	13	C_3
⋮	b	⋮	⋮
⋮	⋮	⋮	⋮

30 C_1
30 C_2
20 C_3

randomly pick 2 features from dataset

max features

this hyperparameter decides, how many features will be tested to find best decision rule for any given node.

$30\,c_1$
$30\,c_2$
$20\,c_3$

root node

randomly selected 2 features = x_1 and x_2

form decision rules based on x_1 and x_2

as x_1 is numerical column

- find all percentiles of x_1

$x_1 = [1.2, 4.5, 3.2, 2, 4, 8, 1.6 \ldots]$

$\downarrow$ percentile values

$P_1 \quad P_2 \quad P_3 \quad P_4 \ldots\ldots P_{100}$

- form 100 decision rules based on 100 percentile values

$$\left.\begin{array}{ll} \text{rule 1} = & x_1 \le P_1 \\ \text{rule 2} = & x_1 \le P_2 \\ \vdots & \\ \text{rule 100} = & x_1 \le P_{100} \end{array}\right\} \text{total 100 decision rules based on } x_1$$

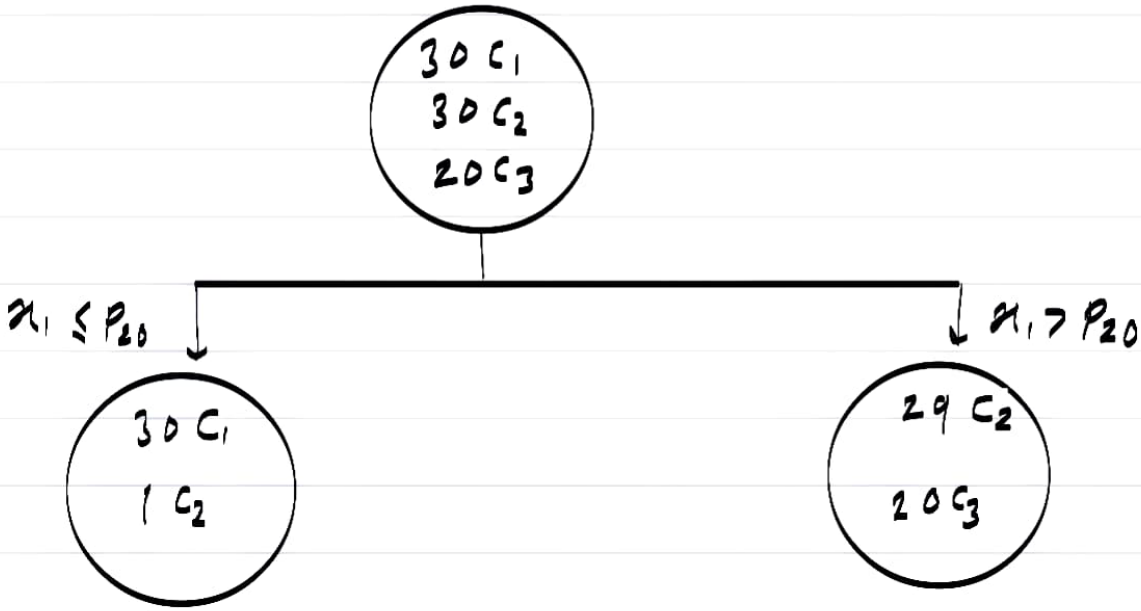
as x_2 is a categorical column

- form 'k' decision rules for 'k' unique categories in x_2

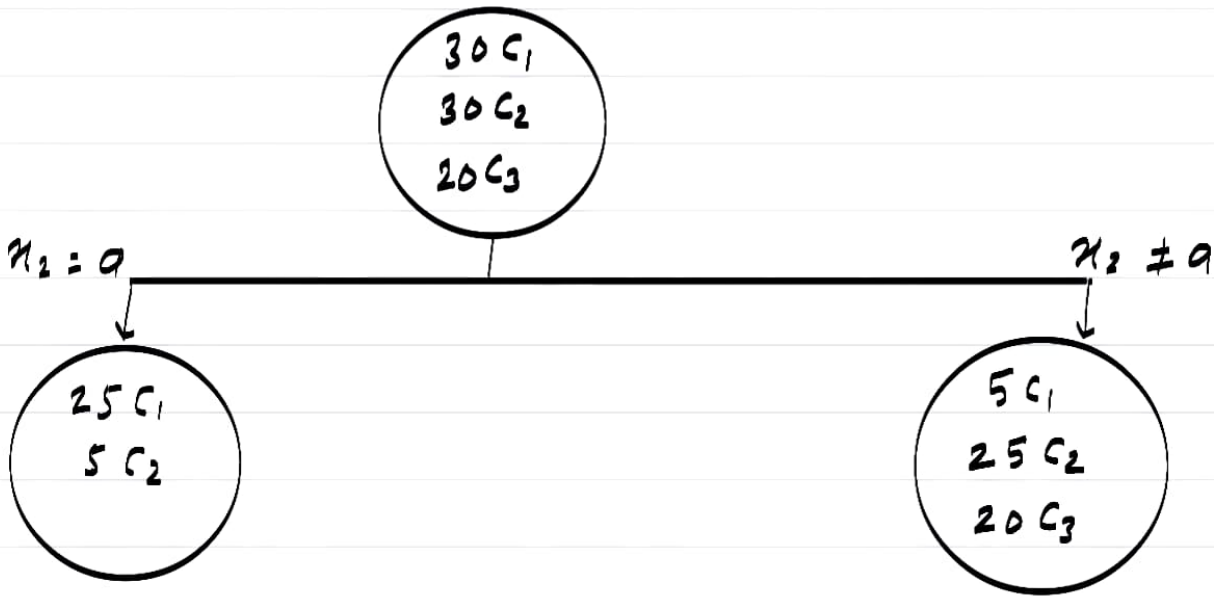
$x_2 = [a, a, b, c, a \ldots]$

$$\left.\begin{array}{ll} \text{rule 1}: & x_2 = a \\ \text{rule 2}: & x_2 = b \\ \text{rule 3}: & x_2 = c \end{array}\right\} \text{total 3 decision rules based on } x_2$$

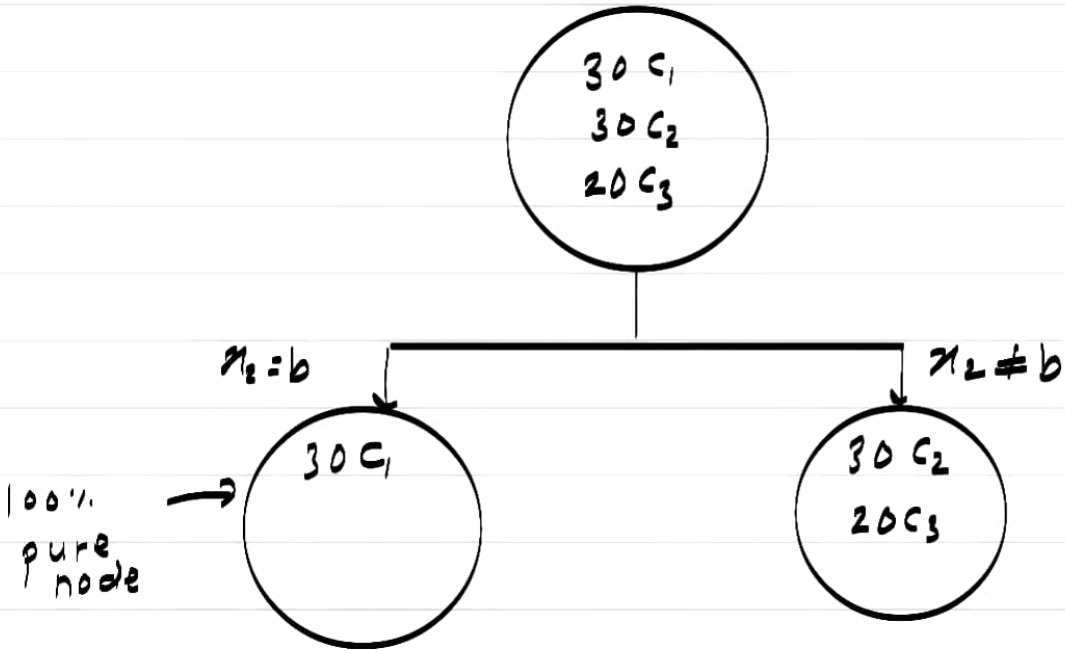
for decision rule based on x_1



for decision rule based on x_2

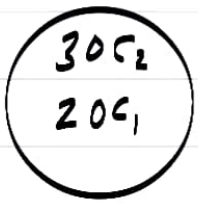


- At 1st level of tree, we'll evaluate 100+3 = 103 decision rules
- we'll find the one decision rule which gives maximum information gain



in the next step, 2 features will be again picked up randomly and best decision rule for next level will be found.

As left child is already 100% pure, it doesn't need further splitting
but right child will need split, similar to its parent node.



randomly pick 2 features and find best decision rule for split.